Day 2 – Arrays (Traversal, Insert, Delete) – Java DSA

1️⃣ Array Traversal

Traversal ka matlab hai **array ke sabhi elements ko ek-ek karke access karna**.

Example Array

10 20 30 40

Index representation

Index	Value
0	10
1	20
2	30
3	40

Java Code – Array Traversal

```
public class Main {

    public static void main(String[] args) {

        int[] arr = {10,20,30,40};

        for(int i = 0; i < arr.length; i++){

            System.out.println(arr[i]);

        }

    }

}
```

Output

```
10
20
30
40
```

Time Complexity

O(n)

2️⃣ Insert Element in Array

Array ka size normally **fixed hota hai**, lekin concept samajhne ke liye hum manually element insert karte hain.

Example

Initial Array

10 20 30 40

Insert **25 at index 2**

Result

10 20 25 30 40

Java Code – Insert Element

```
public class Main {

    public static void main(String[] args) {

        int[] arr = new int[5];

        arr[0] = 10;
        arr[1] = 20;
        arr[2] = 30;
        arr[3] = 40;

        int size = 4;
        int position = 2;
        int value = 25;

        for(int i = size; i > position; i--){

            arr[i] = arr[i-1];

        }

        arr[position] = value;

        size++;

        for(int i = 0; i < size; i++){

            System.out.println(arr[i]);

        }

    }

}
```

Output

10
20
25
30
40

Time Complexity

O(n)

Kyuki elements shift karne padte hain.

3️⃣ Delete Element from Array

Example

Initial Array

10 20 30 40

Delete element at index **1**

Result

10 30 40

Java Code – Delete Element

```
public class Main {

    public static void main(String[] args) {

        int[] arr = {10,20,30,40};
        int size = 4;
        int position = 1;

        for(int i = position; i < size-1; i++){

            arr[i] = arr[i+1];

        }

        size--;

        for(int i = 0; i < size; i++){

            System.out.println(arr[i]);

        }

    }

}
```

Output

10
30
40

Time Complexity

O(n)

Kyuki elements shift hote hain.

4️⃣ Important Interview Points

Array operations complexity

Operation	Complexity
Access	O(1)
Traversal	O(n)
Insert	O(n)
Delete	O(n)